

tf.compat.v1.Dimension Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/Dimension

Represents the value of one dimension in a TensorShape.

Function Signatures

```
tf.compat.v1.Dimension( value )
```

Code Examples

```
tf.compat.v1.Dimension(  
value  
)
```

```
tf.compat.v1.Dimension(  
value  
)
```

```
assert_is_compatible_with(  
other  
)
```

```
assert_is_compatible_with(  
    other  
)
```

```
is_compatible_with(  
    other  
)
```

```
is_compatible_with(  
    other  
)
```

```
merge_with(  
    other  
)
```

```
merge_with(  
    other  
)
```

Documentation

Represents the value of one dimension in a `TensorShape`.

Migrate to TF2

In TF2, members of a `TensorShape` object are integers. The `Dimension` class is not part of TF2's data model.

Please refer to the `TensorShape` section of the migration guide on common code patterns adapting `Dimension` objects to a TF2 syntax.

Description

`## Attributes`

Methods

`## assert_is_compatible_with`

[View source](#)

Raises an exception if other is not compatible with this Dimension.

is_compatible_with

[View source](#)

Returns true if other is compatible with this Dimension.

Two known Dimensions are compatible if they have the same value.

An unknown Dimension is compatible with all other Dimensions.

merge_with

[View source](#)

Returns a Dimension that combines the information in self and other.

Dimensions are combined as follows:

add

[View source](#)

Returns the sum of self and other.

Dimensions are summed as follows:

__bool__

[View source](#)

Equivalent to `bool(self.value)`.

__div__

[View source](#)

This function exists only for backwards compatibility purposes; new code should use `__floordiv__` via the syntax `x // y`. Using `x // y` communicates clearly that the result rounds down, and is forward compatible to Python 3.

__eq__

[View source](#)

Returns true if other has the same known value as this Dimension.

__floordiv__

[View source](#)

Returns the quotient of self and other rounded down.

Dimensions are divided as follows:

__ge__

[View source](#)

Returns True if self is known to be greater than or equal to other.

Dimensions are compared as follows:

__gt__

[View source](#)

Returns True if self is known to be greater than other.

Dimensions are compared as follows:

__le__

[View source](#)

Returns True if self is known to be less than or equal to other.

Dimensions are compared as follows:

__lt__

[View source](#)

Returns True if self is known to be less than other.

Dimensions are compared as follows:

__mod__

[View source](#)

Returns self modulo other.

Dimension modulo are computed as follows:

__mul__

[View source](#)

Returns the product of self and other.

Dimensions are summed as follows:

__ne__

[View source](#)

Returns true if other has a different known value from self.

__radd__

[View source](#)

Returns the sum of other and self.

__rdiv__

[View source](#)

Use `__floordiv__` via `x // y` instead.

This function exists only to have a better error message. Instead of:
TypeError: unsupported operand type(s) for `/:` 'int' and 'Dimension',
this function will explicitly call for usage of `//` instead.

__rfloordiv__

[View source](#)

Returns the quotient of other and self rounded down.

__rmod__

[View source](#)

Returns other modulo self.

`__rmul__`

[View source](#)

Returns the product of self and other.

`__rsub__`

[View source](#)

Returns the subtraction of self from other.

`__rtruediv__`

[View source](#)

Use `__floordiv__` via `x // y` instead.

This function exists only to have a better error message. Instead of:
TypeError: unsupported operand type(s) for /: 'int' and 'Dimension',
this function will explicitly call for usage of `//` instead.

`__sub__`

[View source](#)

Returns the subtraction of other from self.

Dimensions are subtracted as follows:

`__truediv__`

[View source](#)

Use `__floordiv__` via `x // y` instead.

This function exists only to have a better error message. Instead of:
`TypeError: unsupported operand type(s) for /: 'Dimension' and 'int'`,
this function will explicitly call for usage of `//` instead.